Section D - ETL/ELT + Airflow + dbt

Q1. ETL vs ELT Architecture

Tasks

1. Compare ETL vs ELT.
   In ETL data is extracted from sources, transformed before loading and only clean, structured data is stored in the target. Raw data is not stored and the heavy transformations are performed outside the warehouse. Low flexibility and hard to reprocess.
   Whilst in ELT, data is extracted, loaded raw in the data lake and transformations are run inside the target system. Data is loaded quickly and raw data is stored. Computing take place in the warehouse. High flexibility and easier to reprocess.

2. Explain why cloud warehouses prefer ELT.
   Cheap storage. Compute power is large and scalable, making transformations inside the warehouse efficient. Useful for analytics, ML and reporting since ELT allows schema-on-read. Data is available immediately after loading and transformations come later.

3. Design modern ELT pipeline for QuickCart.
   a. Extract from sources (csv files)
   b. Load raw data in the data lake (bronze)
   c. Transformations in the data lake (silver)
   d. Analytics layer (gold)

Q3. Airflow Failure Handling

Tasks

1. Explain idempotency in pipelines.
   Idempotency ensures that no matter how many times you run a job, the same result is produced without creating duplicates or incorrect data. Idempotency is key since in the real-world pipelines can fail due to network issues, timeouts, partial write, etc. Therefore, retries and recovery are fundamental. Idempotency ensures retries don't corrupt data.

2. Explain retry mechanisms.
   Orchestration tools like airflow allows automatic retries on failure in order to handle potential failures. When having a DAG, the retries can be performed in the task that failed instead of in the whole job.

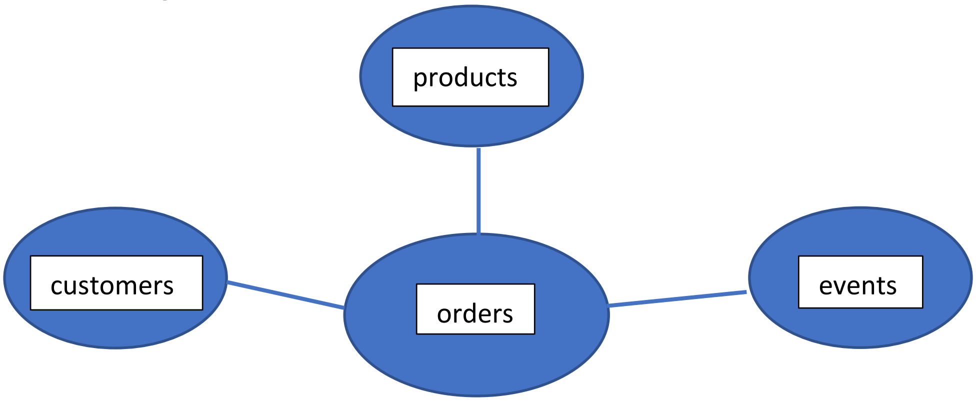
3. How would you prevent duplicate loads?
   Using idempotent operations: overwrite by partition, replace entire table, merge by key, upsert or append with deduplication. We can also use Airflow sensors, ensuring that when the file is extracted, Airflow automatically detects it and loads it.

4. Explain backfilling.
   Backfilling is when the pipeline runs for historical dates that were missed or needs reprocessing. In order to have successful backfilling, the pipeline must be idempotent.

Q5. Data Warehouse Modeling

Tasks

1. Design star schema.



2. Identify: fact tables dimension tables

   Fact table: orders
   Dimension tables: customers, products, events